Objective 1:

To amplify a specific segment of DNA using the Polymerase Chain Reaction (PCR) technique.

Materials and Equipment:

1. DNA template
2. PCR primers (forward and reverse)
3. dNTP mix
4. Taq DNA polymerase
5. PCR buffer
6. $MgCl_2$
7. Nuclease-free water
8. PCR tubes
9. Pipettes and tips
10. Thermal cycler

Procedure:

1. **Preparation of the Master Mix:**
    - The volume of each component needed for the total number of reactions has been calculated, including a small excess to account for pipetting error.
    - The master mix has been prepared by adding the following components in a sterile PCR tube on ice:
        - **PCR buffer (10X):** 2.5 µL
        - **dNTP mix (10 mM each):** 0.5 µL
        - **Forward primer (10 µM):** 1.0 µL
        - **Reverse primer (10 µM):** 1.0 µL
        - **$MgCl_2$ (25 mM):** 1.5 µL (final concentration of 1.5 mM)
        - **Taq DNA polymerase (5 U/µL):** 0.25 µL
        - **Nuclease-free water:** up to 20 µL
    - The mix has been gently mixed by pipetting up and down.
2. **Addition of DNA Template:**
    - 19 µL of the master mix has been aliquoted into each PCR tube.
    - 1 µL of the DNA template (approximately 10-100 ng) has been added to each tube.
    - It has been ensured that the final volume in each tube is 20 µL.
3. **PCR Cycling Conditions:**

- The PCR tubes have been placed in the thermal cycler and the following cycling parameters have been set:

1. **Initial Denaturation:**
    - 95°C for 2 minutes
2. **Amplification (30-35 cycles):**
    - Denaturation: 95°C for 30 seconds
    - Annealing: 50-65°C (depending on primer Tm) for 30 seconds
    - Extension: 72°C for 1 minute per kb of target DNA
3. **Final Extension:**
    - 72°C for 5 minutes
4. **Hold:**
    - 4°C indefinitely

4. **Post-PCR Analysis:**
    - 5 µL of the PCR product has been run on an agarose gel (1-2%) with an appropriate DNA ladder.
    - The gel has been stained with ethidium bromide or a safe alternative (e.g., SYBR Safe) and visualized under UV light.
5. By following these steps, the PCR has been successfully performed, resulting in the amplification of the target DNA sequence.

Notes:

- All reagents have been kept on ice throughout the preparation process.
- Nuclease-free tubes and pipette tips have been used to avoid contamination.
- The annealing temperature has been verified by performing a gradient PCR if the optimal temperature is unknown.
- A negative control (no DNA template) has always been included to check for contamination.

Objective 2: To perform gel electrophoresis of the amplified DNA segment.

Materials and Equipment:

1. Agarose powder
2. 1X TAE buffer (Tris-Acetate-EDTA)
3. Ethidium bromide or SYBR Safe DNA stain
4. Gel casting tray and comb
5. Electrophoresis chamber and power supply
6. Micropipettes and tips
7. PCR-amplified DNA samples
8. DNA ladder (molecular weight marker)
9. Loading dye
10. UV transilluminator

Procedure:

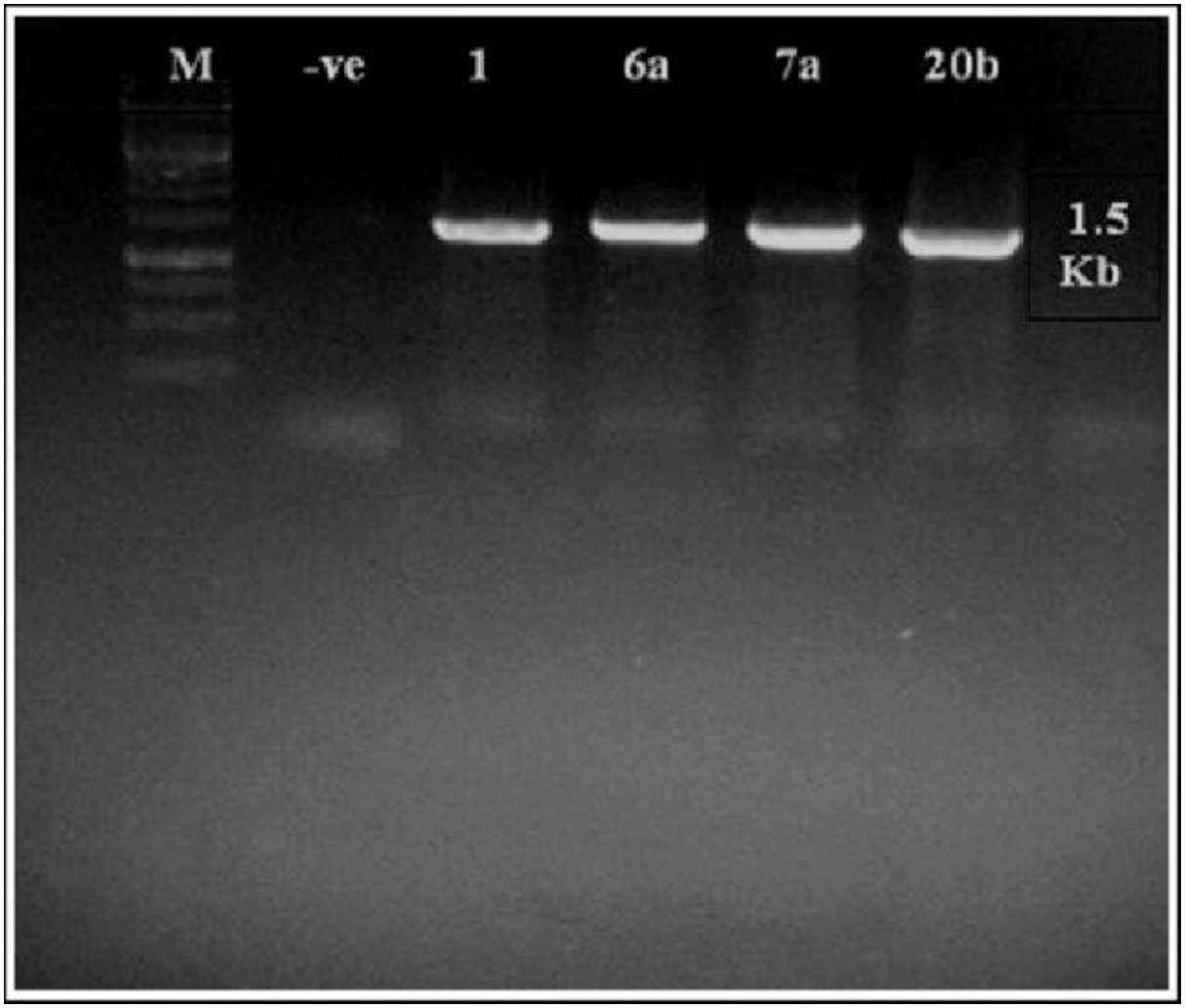
1. **Preparation of Agarose Gel:**
    - The appropriate amount of agarose (typically 1-2% w/v) has been weighed and dissolved in 1X TAE buffer by heating in a microwave until completely melted.
    - The melted agarose solution has been allowed to cool to about 60°C.
    - Ethidium bromide (0.5 µg/mL) or SYBR Safe has been added to the agarose solution for DNA staining.
    - The agarose solution has been poured into the gel casting tray, and a comb has been placed to form wells.
    - The gel has been allowed to solidify at room temperature.
2. **Preparation of Samples:**
    - The PCR-amplified DNA samples have been mixed with loading dye (usually 6X) to a final concentration of 1X.
    - A DNA ladder has been prepared in a similar manner to serve as a molecular weight marker.
3. **Loading the Gel:**
    - The solidified agarose gel has been placed in the electrophoresis chamber, and 1X TAE buffer has been added until the gel is fully submerged.

- The PCR samples and DNA ladder have been carefully loaded into the wells using a micropipette.

4. **Running the Gel:**
    - The electrophoresis chamber has been connected to a power supply, and a voltage of 80-120V has been applied.
    - The gel has been run until the dye front has migrated an appropriate distance (usually 30-45 minutes).

Results:

1. **Amplified DNA Bands:**
    - Clear, distinct bands were observed in the lanes containing the PCR-amplified samples.
    - The size of the amplified DNA fragments corresponded to the expected size of the target DNA segment, confirming successful amplification.
    - No non-specific bands or primer-dimer formations were observed, indicating the specificity of the PCR primers used.
2. **Negative Control:**
    - The negative control (no DNA template) showed no bands, confirming the absence of contamination in the PCR reagents and master mix.
3. **Positive Control:**
    - The positive control (known DNA template) produced a band of the expected size, validating the effectiveness of the PCR conditions and reagents.
4. **Quantification:**
    - The intensity of the bands suggests a high yield of the amplified product, indicating efficient PCR amplification.



Notes:

- Gloves and protective eyewear have been worn throughout the process to handle ethidium bromide or SYBR Safe.
- Care has been taken to avoid overloading the wells to prevent sample spillover.
- The gel has been run slowly to achieve better separation of DNA bands.